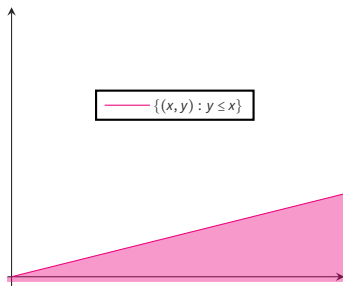


POSITIVITY PROBLEMS AND EQUALITY TESTS

George Kenison



POSITIVITY AND THE HALTING PROBLEM



Does the following loop halt?

$$(x, y) = (2^{-10}, 1);$$

while $y \geq x$ **do**

$$\begin{pmatrix} x \\ y \end{pmatrix} \leftarrow \begin{pmatrix} 2 & 0 \\ \frac{7}{4} & \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix};$$

end while

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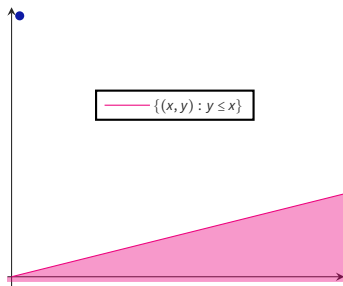
$$M^n \alpha \in \{(x, y) : y \geq x\}, \forall n \in \mathbb{N}?$$

The ‘famous’ halfplane version is commonly phrased in terms of LRSs.

POSITIVITY

Given an integer LRS $\langle u_k \rangle_{k=0}^{\infty}$, decide whether $\forall n \in \mathbb{N}$ s.t. $u_n \geq 0$.

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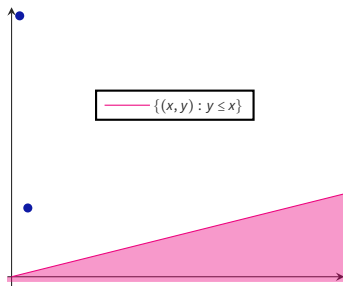
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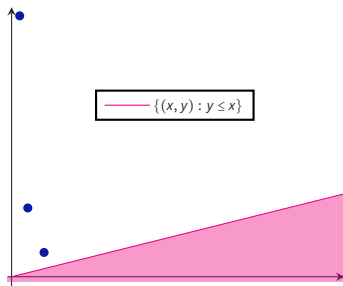
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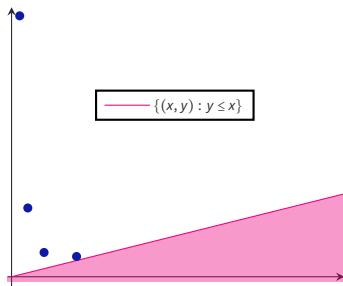
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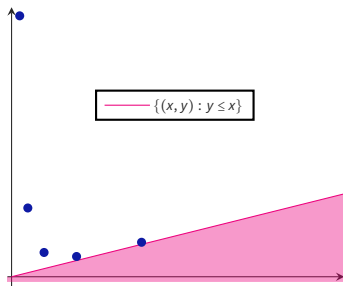
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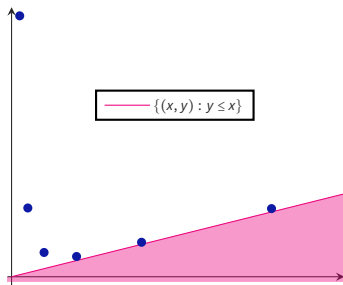
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DECISION PROBLEMS FOR RECURRENCE SEQUENCES

(Are these problems decidable?) [OW15]

Given an LRS $\langle u_n \rangle_{n=0}^{\infty}$ (over $\mathbb{Q}$ or $\mathbb{Z}$)

- Does $u_n = 0$ for some n ?
- Does $u_n = 0$ for infinitely many n ?
- Is $u_n \geq 0$ for all n ?
- Is $u_n \geq 0$ for all but finitely many n ?

An LRS satisfies a rec. relation $u_n = a_1 u_{n-1} + \dots + a_k u_{n-k}$ with initial values $u_0, \dots, u_{k-1}$

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- Does $u_n = 0$ for some n ? **SKOLEM** is open.
- Does $u_n = 0$ for infinitely many n ? **is decidable** (Berstel–Mignotte)
- Is $u_n \geq 0$ for all n ? **POSITIVITY** is open.
- Is $u_n \geq 0$ for all but finitely many n ? **ULTIMATE POSITIVITY** is open.

An LRS satisfies a rec. relation $u_n = a_1 u_{n-1} + \dots + a_k u_{n-k}$ with initial values $u_0, \dots, u_{k-1}$

POSITIVITY RECAP [OW14]

POSITIVITY

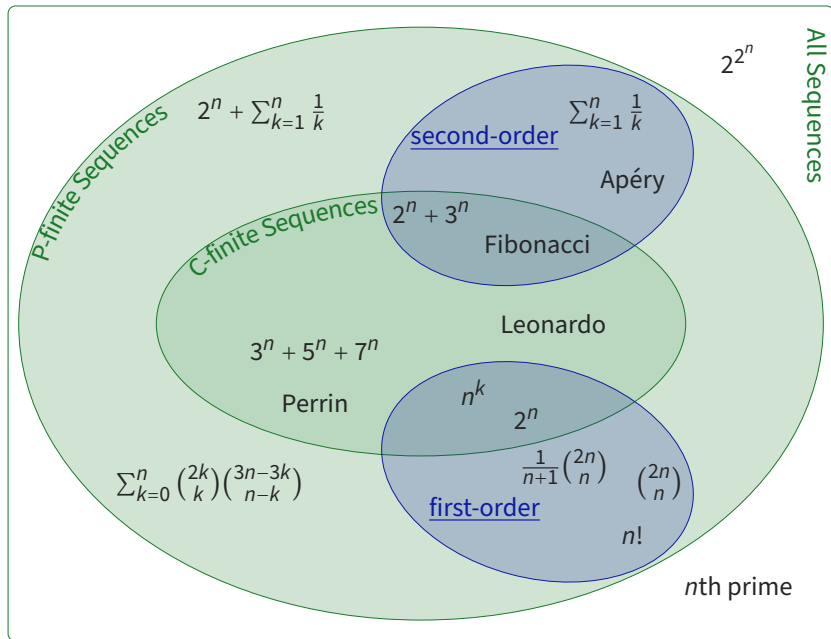
Given a recursively defined seq $\langle u_k \rangle_{k=0}^{\infty}$, decide whether $\forall n \in \mathbb{N}$ s.t. $u_n \geq 0$.

For LRSs (C-finite),

- POSITIVITY is decidable up to order 5.
- If POSITIVITY is decidable at order 6, then we can compute Lagrange constants for a large class of transcendental numbers.

How about a different family of recursively defined sequences?

THE LANDSCAPE OF P-FINITE SEQUENCES [KP11]



POSITIVITY FOR SECOND-ORDER P-FINITE SEQUENCES

- [Ken+] G. Kenison, O. Klurman, E. Lefauchaux, F. Luca, P. Moree, E. C. Sertöz, J. Ouaknine, A. Whiteland, and J. Worrell. *On the Positivity Problem for second-order holonomic sequences*.
- [Ken+21] G. Kenison, O. Klurman, E. Lefauchaux, F. Luca, P. Moree, J. Ouaknine, M. A. Whiteland, and J. Worrell. “On Positivity and Minimality for Second-Order Holonomic Sequences”. In: *International Symposium on Mathematical Foundations of Computer Science, MFCS 2021*. 2021, 67:1–67:15.

Many more papers (e.g., [HK25; IS24; Ken24; KP10]) consider POSITIVITY and related problems in the P-finite (not necessarily second-order) setting.

SECOND-ORDER P-FINITE SEQUENCES

rational sequences $\langle u_n \rangle_{n=-1}^{\infty}$ that satisfy relations of the form

$$p_3(n)u_n = p_2(n)u_{n-1} + p_1(n)u_{n-2} \quad n = 1, 2, 3, \dots$$

where each $p_i \in \mathbb{Q}[x]$, $p_1(n), p_3(n) \neq 0$ for all $n \in \mathbb{N}$.

The solution space of the recurrence relation is a 2-dim vector space

- A soln $\langle u_n \rangle_n$ is uniquely determined by the pair u_{-1} and u_0 .
- $\langle 0 \rangle_n$ is a soln.
- if $\langle u_n \rangle_n$ is a soln, then so is $\alpha \langle u_n \rangle_n = \langle \alpha u_n \rangle_n$ for every $\alpha \in \mathbb{R}$.
- if $\langle w_n \rangle_n$ and $\langle v_n \rangle_n$ are solutions, then so is $\langle w_n \rangle_n + \langle v_n \rangle_n = \langle v_n + w_n \rangle_n$.

SECOND-ORDER P-FINITE SEQUENCES

rational sequences $\langle u_n \rangle_{n=-1}^{\infty}$ that satisfy relations of the form

$$u_n = b_n u_{n-1} + a_n u_{n-2} \quad \text{for } n = 1, 2, 3, \dots$$

with $a_n := \frac{p_1(n)}{p_3(n)}$ and $b_n := \frac{p_2(n)}{p_3(n)}$.

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DECISION PROBLEMS

Definition (Minimal solutions)

A soln $\langle u_n \rangle_n$ to $p_3(n)u_n = p_2(n)u_{n-1} + p_1(n)u_{n-2}$ is *minimal* if $\langle u_n \rangle_n$ is non-trivial and there exists a second soln $\langle v_n \rangle_n$ such that $\lim_{n \rightarrow \infty} u_n/v_n = 0$.

Decision Problems for Recurrence Relations

- Does a given recurrence relation admit Positive solutions?
- Does a given recurrence relation admit Minimal solutions?

Decision Problems for Recurrence Sequences

- (POSITIVITY) Is a given sequence $\langle u_n \rangle_n$ Positive? (i.e., $u_n \geq 0$, $\forall n$)
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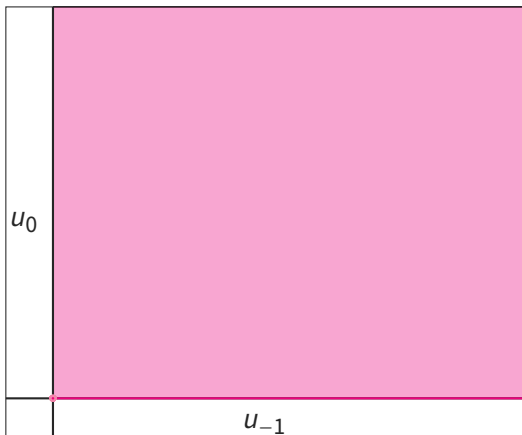
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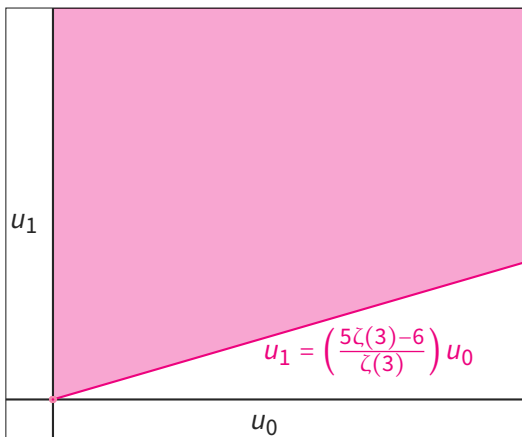
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POSITIVE SOLUTIONS TO $u_{n+1} = u_n + u_{n-1}$



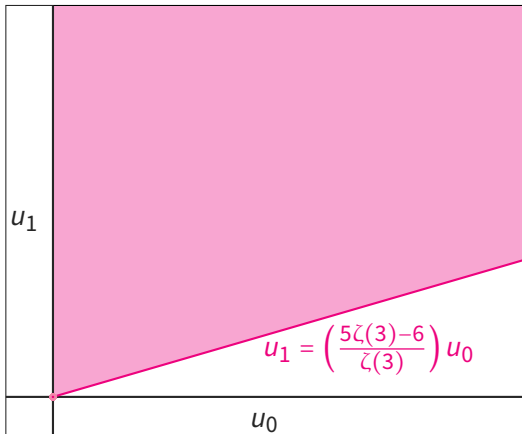
Set $I = \{(u_{-1}, u_0) : u_n \geq 0, \forall n\}$ of initial values that determine a positive solution are closed under linear combinations with positive coefficients, so I is a convex linear cone.

POSITIVE SOLUTIONS TO $n^3 u_{n+1} = (34n^3 - 51n^2 + 27n - 5)u_n - (n-1)^3 u_{n-1}$



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$$\langle A_n \rangle_{n=0}^{\infty} = \langle 1, 5, 73, 1445, \dots \rangle, \quad \langle B_n \rangle_{n=0}^{\infty} = \langle 0, 6, 351/4, 62531/36, \dots \rangle.$$

Because $\frac{B_n - \zeta(3)A_n}{A_n} \rightarrow 0$ [Apé79; Poo79], $\alpha \langle \zeta(3)A_n - B_n \rangle_n$ is a *minimal* soln (substitute initial values!).

THE CONTINUED FRACTION CONNECTION

$$\mathbf{K} \frac{a_n}{b_n} := \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{a_3 + \ddots + \frac{b_n}{a_n}}}}$$

Theorem (Pincherle 1894 [])

- $\mathbf{K}(a_n/b_n)$ converges in $\hat{\mathbb{R}}$ if and only if $u_n = b_n u_{n-1} + a_n u_{n-2}$ admits minimal solutions.
- If $\langle w_n \rangle_{n=-1}^{\infty}$ is a minimal solution of $u_n = b_n u_{n-1} + a_n u_{n-2}$ then $\mathbf{K}(a_n/b_n)$ converges to $-u_0/u_{-1} \in \hat{\mathbb{R}}$. (MINIMALITY !)

THE CONTINUED FRACTION CONNECTION

$$\mathbf{K}_{n=1}^{\infty} \frac{a_n}{b_n} := \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{a_3 + \ddots + \frac{b_n}{a_n}}}}$$

$$\phi = 1 + \mathbf{K}_{n=1}^{\infty} \frac{1}{1},$$

$$\frac{4}{\pi} = 1 + \mathbf{K}_{n=1}^{\infty} \frac{(2n-1)^2}{2},$$

$$\frac{5\zeta(3) - 6}{\zeta(3)} = \mathbf{K}_{n=1}^{\infty} \frac{-n^6}{34n^3 + 51n^2 + 27n + 5}.$$

Theorem ([Ken+21])

Given a second-order P-finite recurrence relation

$p_3(n)u_n = p_2(n)u_{n-1} + p_1(n)u_{n-2}$. We have the following:

- 1. There is a convex linear cone such that all terms of a solution $\langle u_n \rangle_n$ are non-negative if and only if u_0/u_{-1} lies in the cone.*
- 2. We can decide POSITIVITY except when u_0/u_{-1} lies on the boundary of the cone.*

Further, for the class of second-order P-finite sequences, the problem of deciding POSITIVITY Turing-reduces to that of MINIMALITY (i.e., a procedure to decide MINIMALITY implies the existence of a procedure to decide POSITIVITY).

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REFERENCES [1]

- [Apé79] R. Apéry. “Irrationalité de $\zeta(2)$ et $\zeta(3)$ ”. In: *Astérisque* 61 (1979). Luminy Conference on Arithmetic, pp. 11–13.
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- [Ken+] G. Kenison, O. Klurman, E. Lefauchaux, F. Luca, P. Moree, E. C. Sertöz, J. Ouaknine, A. Whiteland, and J. Worrell. *On the Positivity Problem for second-order holonomic sequences*.

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- [OW15] J. Ouaknine and J. Worrell. “On Linear Recurrence Sequences and Loop Termination”. In: *ACM SIGLOG News* 2.2 (2015), pp. 4–13.
- [Poo79] A. van der Poorten. “A proof that Euler missed ...: An informal report”. In: *The Mathematical Intelligencer* 1.4 (1979), pp. 195–203.

Sequences obeying relations of the form

$$(\alpha_1 n + \alpha_0)u_n = (\beta_1 n + \beta_0)u_{n-1} + (\gamma_1 n + \gamma_0)u_{n-2}$$

are the *second-order degree-one P-finite sequences*.

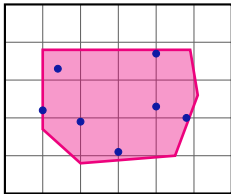
Theorem ([Ken+])

POSITIVITY and MINIMALITY are decidable for the class of second-order degree-one P-finite sequences with two distinct characteristic roots (i.e., associated $p(x) = \alpha_1 x^2 - \beta_1 x - \gamma_1$ has two distinct rational roots).

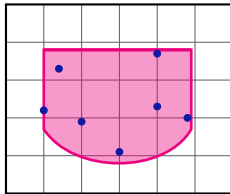
Under these restrictive conditions, the equality checks performed as part of the decision procedure make use of the following recent results:

1. **[FR22]**, for determining all polynomial relations of a given family of E-functions evaluated at a common algebraic point.
2. **[SOW]**, for determining all linear relations of 1-periods.

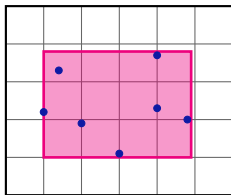
INVARIANTS AND REACHABLE STATES



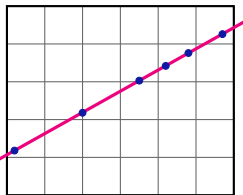
Polyhedrons



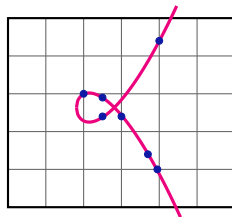
Semialgebraic sets



Intervals



Affine/linear sets



Algebraic sets

An **invariant** is a set that overapproximates the set of **reachable states** $\mathcal{O}_L$.

An **algebraic set** is defined by the common zeroes of a family of polynomials.